Spectral Contraction and T-Independence in Barrington Branching Programs: Resolving the Uniform Mixing Conjecture for Point Functions over S_5

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Abstract

We study the distribution D_T over the symmetric group S_5 induced by evaluating a Barrington branching program for the point function CC_T on uniformly random inconsistent inputs. We prove two results. First, $D_T = D_{T'}$ for all $T, T' \in \{0,1\}^\ell$ (exact T-independence), via a combinatorial flip bijection that requires no properties of the underlying group. Second, $d_{\mathrm{TV}}(D_T, \mathrm{Uniform}(S_5)) \leq \mathrm{negl}(\ell)$ (spectral contraction), via Fourier analysis on S_5 using the Diaconis–Shahshahani Upper Bound Lemma. The contraction proof proceeds by exact computation of Fourier norms for the standard representation $(4,1)$ at the first two levels of the commutator tree, followed by an algebraic proof that the operator norm drops below 1 at the third level — enabled by the discovery that the level-2 obstruction has a regular simplex structure in $\mathbb{R}^4$ that the commutator necessarily breaks. The direction-mixing mechanism at level 3 is proved via the solvability of $S_4 \cong \mathrm{Stab}(k)$: the derived series $S_4 \rhd A_4 \rhd V_4 \rhd \{e\}$ forces the level-3 commutator into the abelian Klein four-group, making it trivial within any single stabilizer. We further prove that this spectral contraction mechanism is *specific to* S_5: for S_n with $n \geq 6$, a dimensional obstruction (the preserved subspace of D_1 has dimension $n-3$, and two such subspaces in $\mathbb{R}^{n-1}$ must intersect when $n \geq 6$) prevents operator-norm contraction at level 3. Together, these results resolve the Uniform T-Independence Conjecture for Barrington point-function branching programs, establish an information-theoretic foundation for the security of branching-program-based obfuscation, and characterize S_5 as the unique symmetric group where level-3 spectral contraction occurs.

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1 Introduction

Indistinguishability obfuscation (iO) for general circuits is one of the most powerful primitives in cryptography, implying a vast array of applications from functional encryption to deniable encryption [10, 18]. The first candidate constructions [10] were based on matrix branching programs over multilinear maps, randomized via Kilian's technique [15], following the classical correspondence established by Barrington's theorem [1]: every NC^1 circuit of depth d can be simulated by a branching program of width 5 and length at most 4^d over the symmetric group S_5.

The security of such constructions depends critically on the algebraic properties of the distributions that arise when branching programs are evaluated on "incorrect" inputs. Specifically, for a point function $\mathrm{CC}_T : \{0,1\}^\ell \to \{0,1\}$ that outputs 1 only on $x = T$, the Barrington program BP_T produces a designated permutation $\pi_{\text{accept}} \in S_5$ on input T and the identity e on all other inputs. When the program is evaluated on a random bit string $b \in \{0,1\}^L$ that is *inconsistent* (at least one variable receives different values at different steps), the output is a random element of S_5 whose distribution we denote D_T.

Two properties of D_T are essential for security:

(i) **T-independence:** $D_T = D_{T'}$ for all T, T', so the distribution reveals nothing about which target was chosen.

(ii) **Uniform mixing:** $d_{\mathrm{TV}}(D_T, \mathrm{Uniform}(S_5)) \leq \mathrm{negl}(\ell)$, so the distribution is indistinguishable from random.

These properties were widely assumed but not formally established: the distributional behavior of Barrington programs on inconsistent inputs — a foundational question open since 1989 — was implicitly relied upon in security analyses without proof. Property (i) was used without justification, and property (ii) was supported by heuristic arguments and limited computational evidence.

Our contributions. We prove both properties.

For T-independence, we give a short combinatorial proof (Theorem 3.1) via a flip bijection F_S that maps inconsistent paths for T to inconsistent paths for T' while preserving the output permutation. The proof uses no property of S_5 and extends to conjunctions with wildcards.

For uniform mixing, we develop a Fourier-analytic framework on S_5 (Section 4). Using the Diaconis–Shahshahani Upper Bound Lemma, we reduce the problem to bounding the Fourier norms of D_T in each irreducible representation. The standard representation $(4, 1)$ of dimension 4 is the bottleneck. We establish contraction at the first two levels of the commutator tree by exact rational computation (Theorems 4.4 and 4.5), discover that the level-2 obstruction has a regular simplex structure (Theorem 4.6), and prove that this structure is necessarily broken at level 3 (Theorem 4.10), yielding operator-norm contraction with $\lambda \leq 0.0354$. An inductive argument then gives super-exponential decay for all higher levels (Theorem 4.11).

Implications. Combined, our results show that for $\ell \geq 8$, the distribution over inconsistent evaluations of BP_T is simultaneously independent of T and negligibly close to uniform. This provides an information-theoretic foundation for the security of obfuscation schemes based on Barrington branching programs for point functions and, more broadly, for conjunctions with bounded wildcard density.

Related work. The first iO candidate from branching programs [10] was followed by constructions for specific function classes: lockable obfuscation for point functions from LWE [11, 12], conjunction obfuscation from noisy encodings and LPN [5, 6], and entropic Ring-LWE approaches [4]. The landmark result of Jain, Lin, and Sahai [14] achieved iO for general circuits from well-founded assumptions, departing from the branching-program paradigm. Recent lattice-based candidates [3, 19, 13] use matrix groups $\mathrm{GL}_n(\mathbb{Z}_q)$ rather than S_5, while impossibility results [9] constrain the space of viable assumptions. For these constructions, the mixed-input security model — where an adversary evaluates the program on inconsistent inputs — remains central [8].

Fourier-analytic methods for branching programs have a separate tradition in the pseudorandomness literature [17, 16], but that work uses Fourier analysis on $\{0,1\}^n$ (the input space) to fool Boolean outputs of read-once programs. Our analysis instead uses Fourier analysis on S_5 (the output group) to bound total variation of the group-valued output distribution of multi-read programs — a fundamentally different analytic object requiring different tools. We are not aware of prior work connecting the Diaconis–Shahshahani framework [7] to branching-program security.

Techniques. The proof of T-independence is purely combinatorial. The proof of spectral contraction uses three different methods at three different scales: exact symbolic algebra (levels 0–2), an algebraic argument via the solvability of S_4 (level 3), and analytic induction (levels 4+). The key discovery that bridges levels 2 and 3 is the simplex structure of the preserved directions: the 270 out of 405 level-2 configurations with operator norm 1 each preserve one of exactly 5 directions in $\mathbb{R}^4$, forming a regular simplex with pairwise $|\cos\theta| = 1/4$. An algebraic proof shows that valid level-3 configurations cannot share a preserved direction: within the stabilizer $\mathrm{Stab}(k) \cong S_4$, the derived series terminates at depth 3, forcing the commutator to be trivial. A separate analysis of S_n for $n \geq 6$ reveals that this mechanism is unique to S_5: the level-1 Fourier transform preserves a subspace of dimension $n-3$ in $\mathbb{R}^{n-1}$, and the intersection bound $2(n-3)-(n-1) = n-5 \geq 1$ blocks contraction for all $n \geq 6$.

2 Preliminaries

2.1 Branching Programs and Barrington's Theorem

Definition 2.1 (Permutation Branching Program)**.** *A* permutation branching program *of width w and length L over a group $G \leq S_w$ is a sequence of instructions $\Pi = ((i_1, g_1^{(0)}, g_1^{(1)}), \ldots, (i_L, g_L^{(0)}, g_L^{(1)}))$, where each $i_j \in [\ell]$ is a variable index and $g_j^{(0)}, g_j^{(1)} \in G$. On input $x \in \{0,1\}^\ell$, the program evaluates $\Pi(x) = \prod_{j=1}^{L} g_j^{(x_{i_j})}$. The program* α-computes *a function f if $\Pi(x) = \alpha$ when $f(x) = 1$ and $\Pi(x) = e$ when $f(x) = 0$.*

Theorem 2.2 (Barrington [1])**.** *Every Boolean function computed by a circuit of depth d can be α-computed by a permutation branching program of width 5 and length at most 4^d over S_5, for some $\alpha \neq e$.*

The construction uses the *commutator trick*: given programs P and Q that α-compute f and β-compute g respectively, the concatenation $P \cdot Q \cdot P^{-1} \cdot Q^{-1}$ computes $f \wedge g$ via the group commutator $[\alpha, \beta] = \alpha\beta\alpha^{-1}\beta^{-1}$. This works because S_5 is non-solvable: iterated commutators of non-identity elements remain non-identity.

Definition 2.3 (Point Function)**.** *The point function $\mathrm{CC}_T : \{0,1\}^\ell \to \{0,1\}$ is defined by $\mathrm{CC}_T(x) = 1$ if and only if $x = T$.*

For point functions, the Barrington program has ℓ leaves (one per bit position), each comparing x_i to T_i. Leaf i outputs a fixed transposition σ_i when $x_i = T_i$ and the identity e when $x_i \neq T_i$. The tree of commutators has depth $\log_2 \ell$, yielding a program of length $L = 4^{\log_2 \ell} = \ell^2$.

2.2 The Distribution D_T

Definition 2.4 (Inconsistent Path)**.** *A bit string $b = (b_1, \ldots, b_L) \in \{0,1\}^L$ is* inconsistent *with respect to the program Π if there exist steps j, j' reading the same variable ($i_j = i_{j'}$) with $b_j \neq b_{j'}$.*

Definition 2.5 (D_T)**.** *Let $\mathcal{I} \subset \{0,1\}^L$ be the set of inconsistent paths. The distribution D_T over S_5 is defined by*

$$D_T(\pi) = \Pr_{b \sim \mathrm{Uniform}(\mathcal{I})} \left[\prod_{j=1}^{L} g_j^{(b_j)} = \pi \right].$$

2.3 Representation Theory of S_5

The symmetric group S_5 has 7 irreducible representations, labeled by partitions of 5:

Partition λ	Dimension d_λ	χ_λ(transposition)	Name
(5)	1	1	Trivial
$(4,1)$	4	2	Standard
$(3,2)$	5	1	—
$(3,1,1)$	6	0	—
$(2,2,1)$	5	-1	Sign $\otimes$ $(3,2)$
$(2,1,1,1)$	4	-2	Sign $\otimes$ Standard
(1^5)	1	-1	Sign

The dimensional check confirms: $1^2 + 4^2 + 5^2 + 6^2 + 5^2 + 4^2 + 1^2 = 120 = |S_5|$.

Definition 2.6 (Standard Representation)**.** *The standard representation $\rho_{(4,1)} : S_5 \to \mathrm{GL}(\mathbb{R}^4)$ acts on the hyperplane $V = \{v \in \mathbb{R}^5 : \sum_i v_i = 0\}$ by permuting coordinates. We use the orthonormal basis*

$$f_1 = (e_0 - e_1)/\sqrt{2}, \quad f_2 = (e_0 + e_1 - 2e_2)/\sqrt{6},$$
$$f_3 = (e_0 + e_1 + e_2 - 3e_3)/\sqrt{12}, \quad f_4 = (e_0 + e_1 + e_2 + e_3 - 4e_4)/\sqrt{20},$$

so that $\rho_{(4,1)}(\sigma)$ is orthogonal for all $\sigma \in S_5$.

2.4 The Diaconis–Shahshahani Upper Bound Lemma

Theorem 2.7 (Diaconis–Shahshahani [7])**.** *For any probability distribution Q on a finite group G,*

$$4 \cdot d_{\mathrm{TV}}(Q, U)^2 \leq \sum_{\rho \neq \mathrm{trivial}} d_\rho \cdot \|\widehat{Q}(\rho)\|_{\mathrm{F}}^2,$$

where the sum is over non-trivial irreducible representations ρ of dimension d_ρ, $\widehat{Q}(\rho) = \sum_{g \in G} Q(g)\rho(g)$ is the Fourier transform, and $\|\cdot\|_{\mathrm{F}}$ denotes the Frobenius norm.

For S_5, the standard representation $(4,1)$ is the *mixing bottleneck*. Although $(2,1,1,1) = \text{sign} \otimes (4,1)$ shares the same normalized character ratio $|\chi(\sigma)|/d_\rho = 1/2$, its leaf projection $\frac{1}{2}(I - \rho_{(4,1)}(\sigma))$ has rank 1 (versus rank 3 for $(4,1)$), causing it to contract at level 1 with operator norm $\leq (1/2)^3 = 1/8$. The remaining representations satisfy $|\chi(\sigma)|/d_\rho \leq 1/5$ and likewise contract earlier (Theorem 4.2). It therefore suffices to bound $\|\widehat{D}(4,1)\|_{\mathrm{F}}^2$.

3 Exact T-Independence

Theorem 3.1 (T-Independence)**.** *Let* BP_T *be the Barrington branching program for the point function* CC_T *over* $\{0,1\}^\ell$*. Then* $D_T = D_{T'}$ *for all* $T, T' \in \{0,1\}^\ell$*.*

Proof. Let $S = \{i : T_i \neq T'_i\}$ be the set of differing bit positions. Define $F_S : \{0,1\}^L \to \{0,1\}^L$ by

$$F_S(b)_j = \begin{cases} 1 - b_j & \text{if } i_j \in S, \\ b_j & \text{otherwise,} \end{cases}$$

where i_j is the variable read at step j.

We verify three properties:

(1) F_S **is an involution.** Applying F_S twice flips each bit in S-positions twice, returning to the original string. Hence $F_S \circ F_S = \mathrm{Id}$.

(2) F_S **preserves inconsistency.** F_S flips *all* readings of each variable x_i with $i \in S$ simultaneously. If all readings of x_i were equal (say, all 0), after flipping they are all 1 — still consistent. If some readings differed, after flipping they still differ. Variables with $i \notin S$ are unaffected. Therefore inconsistency of each variable is preserved individually.

(3) $\varphi_{T'}(b) = \varphi_T(F_S(b))$ **for all** b**.** At each step j reading variable x_{i_j}:

- If $i_j \notin S$: $T_{i_j} = T'_{i_j}$ and F_S does not flip b_j. Same input, same target bit, same output.
- If $i_j \in S$: $T'_{i_j} = 1 - T_{i_j}$ and F_S flips b_j. The swap of target bit and input bit cancel: $g_j^{(b_j)}$ under T' equals $g_j^{(1-b_j)}$ under T, which equals $g_j^{(F_S(b)_j)}$ under T.

Hence the per-step outputs, and their product, are equal.

Combining (1)–(3):

$$\begin{aligned} D_{T'}(\pi) &= \Pr[\varphi_{T'}(b) = \pi \mid b \in \mathcal{I}] \\ &= \Pr[\varphi_T(F_S(b)) = \pi \mid b \in \mathcal{I}] && \text{(by (3))} \\ &= \Pr[\varphi_T(b') = \pi \mid F_S^{-1}(b') \in \mathcal{I}] && \text{(substituting } b' = F_S(b)) \\ &= \Pr[\varphi_T(b') = \pi \mid b' \in \mathcal{I}] && \text{(by (2), since } F_S = F_S^{-1} \text{ by (1))} \\ &= D_T(\pi). && \square \end{aligned}$$

Remark 3.2. *The proof uses no property of* S_5 *— it works for branching programs of any width over any group. It applies to all programs for point functions, not just Barrington's construction.*

Remark 3.3. *The proof extends to conjunctions with wildcards: if position* i *is a wildcard, the corresponding leaf outputs the same permutation* σ_i *for both bit values. Flipping* b_j *at wildcard positions changes both branches identically, so the output is unchanged. The bijection* F_S *is restricted to non-wildcard differing positions.*

Remark 3.4 (Unconditional T-independence)**.** *The bijection* F_S *preserves inconsistency of each variable individually (property (2)), and therefore also preserves the set of* consistent *paths. Consequently,* $D_T = D_{T'}$ *holds not only for the distribution conditioned on inconsistent paths, but also for*

the unconditional distribution over all 2^L paths and for the distribution over consistent paths alone. The conditioning on inconsistency is needed for the security argument (consistent paths reveal T), not for T-independence itself.

Remark 3.5 (Variable-grouping invariance)**.** *The distribution D_T depends on the choice of transpositions assigned to each leaf, but* not *on how variables are grouped into the binary commutator tree. Exhaustive computation for $\ell = 4$ confirms that all three possible balanced groupings of 4 variables yield identical distributions. This is because F_S acts on variables, not on tree positions.*

4 Spectral Contraction

By Theorem 3.1, D_T is independent of T, so it suffices to show $d_{\mathrm{TV}}(D_T, U) \leq \mathrm{negl}(\ell)$ for any fixed T. By the Diaconis–Shahshahani bound (Theorem 2.7), this reduces to bounding $\|\widehat{D}(\rho)\|_{\mathrm{F}}^2$ for each non-trivial representation ρ.

Remark 4.1 (Sign representation vanishes identically)**.** *For the sign representation (1^5), the leaf Fourier transform is $\widehat{D}(\mathrm{sign}) = \frac{1}{2}(\mathrm{sign}(\sigma) + 1) = 0$ for any transposition σ, since $\mathrm{sign}(\sigma) = -1$. This propagates to all levels by multiplicativity, so the sign representation contributes nothing to the Diaconis–Shahshahani bound. The sum thus runs over 5 (not 6) non-trivial representations.*

Proposition 4.2 (Contraction in the remaining representations)**.** *For all five non-trivial, non-sign representations ρ of S_5, the operator norm $\|\widehat{D}_3(\rho)\|_{\mathrm{op}}$ is strictly less than 1 for every valid level-3 configuration. The representation $(4,1)$ is the unique bottleneck:*

ρ	d_ρ	*Leaf rank*	$\max\|\widehat{D}_3(\rho)\|_{\mathrm{op}}$	*First level* < 1
$(4,1)$	*4*	*3*	3.54×10^{-2}	*Level* 3
$(3,2)$	*5*	*3*	1.77×10^{-5}	*Level* 2
$(3,1,1)$	*6*	*3*	1.05×10^{-5}	*Level* 2
$(2,2,1)$	*5*	*2*	$< 10^{-10}$	*Level* 1
$(2,1,1,1)$	*4*	*1*	$< 10^{-10}$	*Level* 1

The representations $(2,1,1,1) = \mathrm{sign} \otimes (4,1)$ and $(2,2,1) = \mathrm{sign} \otimes (3,2)$ have leaf projections of rank 1 and 2 respectively. Their level-1 operator norms are at most $(1/2)^3 = 1/8$, so they contract immediately. The representations $(3,2)$ and $(3,1,1)$ have rank-3 leaf projections like $(4,1)$, but lack the simplex obstruction (Theorem 4.6): all 405 level-2 quadruples have operator norm < 0.085, so they contract one level earlier. Verified exhaustively over all 73810 level-3 configurations for each representation *(Script 13 in the accompanying code).*

We focus on the bottleneck representation $(4,1)$.

4.1 Framework: Projections and the Fourier Domain

For a transposition $\sigma = (a\ b) \in S_5$, the standard representation $\rho_{(4,1)}(\sigma)$ has eigenvalues $\{+1, +1, +1, -1\}$ (three fixed directions and one flipped). The Fourier transform of the leaf distribution $\{\sigma : 1/2,\ e : 1/2\}$ is

$$P_\sigma := \frac{1}{2}\big(\rho_{(4,1)}(\sigma) + I_4\big),$$

which is a rank-3 orthogonal projection (idempotent, symmetric, trace 3).

Lemma 4.3. *For any transposition $\sigma \in S_5$:*

(a) $P_\sigma^2 = P_\sigma$ *(idempotent),*
(b) $P_\sigma^\top = P_\sigma$ *(symmetric),*
(c) $\mathrm{rank}(P_\sigma) = 3$*,*
(d) $\|P_\sigma\|_{\mathrm{F}}^2 = \mathrm{tr}(P_\sigma) = 3$*.*

The null space of P_σ is spanned by the projection of $e_a - e_b$ onto the hyperplane V, where $\sigma = (a\ b)$.

At each level of the Barrington commutator tree, the Fourier transform is a product of projections. For the single AND block at level 1 using transpositions σ, τ (with $\sigma^{-1} = \sigma$, $\tau^{-1} = \tau$ since they are transpositions):

$$\widehat{D}_1 = P_\sigma \cdot P_\tau \cdot P_\sigma \cdot P_\tau = (P_\sigma P_\tau)^2.$$

At level 2, using two level-1 blocks A (with σ_1, τ_1) and B (with σ_2, τ_2):

$$\widehat{D}_2 = (P_{\sigma_1} P_{\tau_1})^2 \cdot (P_{\sigma_2} P_{\tau_2})^2 \cdot (P_{\tau_1} P_{\sigma_1})^2 \cdot (P_{\tau_2} P_{\sigma_2})^2,$$

where the inverse blocks reverse the transposition order.

4.2 Theorem 1: Level 0 to Level 1 Contraction

Theorem 4.4 (Universal Level-1 Contraction)**.** *For any pair of adjacent transpositions* (σ, τ) *in* S_5 *(i.e.,* $[\sigma, \tau] \neq e$*):*

$$\|(P_\sigma P_\tau)^2\|_{\mathrm{F}}^2 = \frac{129}{64}.$$

The contraction ratio is $\frac{129/64}{3} = \frac{43}{64} \approx 0.6719$*, and is* ***universal*** *— identical for all* 30 *adjacent transposition pairs.*

Proof. There are $\binom{5}{2} = 10$ transpositions in S_5, forming 30 adjacent pairs (pairs sharing exactly one element, equivalently $[\sigma, \tau] \neq e$). For each pair, we compute $(P_\sigma P_\tau)^2$ in the orthonormal basis of Section 2 using exact rational arithmetic, and verify $\|(P_\sigma P_\tau)^2\|_{\mathrm{F}}^2 = 129/64$.

The universality follows from the geometry of the standard representation: the null vectors of P_σ and P_τ (projections of $e_a - e_b$ and $e_c - e_d$ onto V) have inner product $\pm 1/2$ for all adjacent pairs, determined by the simplex geometry of $\{e_0, \ldots, e_4\}$ in V. Since $\|(P_\sigma P_\tau)^2\|_{\mathrm{F}}^2$ depends only on this angle, it is universal. □

4.3 Theorem 2: Level 1 to Level 2 Contraction

Theorem 4.5 (Level-2 Contraction)**.** *For any valid quadruple of transpositions* $(\sigma_1, \tau_1, \sigma_2, \tau_2)$ *with* $[\sigma_i, \tau_i] \neq e$ *and* $[[\sigma_1, \tau_1], [\sigma_2, \tau_2]] \neq e$*:*

$$\|\widehat{D}_2\|_{\mathrm{F}}^2 \leq \frac{67586345}{67108864} \approx 1.0071.$$

The contraction ratio relative to level 1 *is at most* $\frac{67586345/67108864}{129/64} \approx 0.4997 < 1$*.*

Proof. We enumerate all 405 valid quadruples (from the 30 adjacent pairs with the additional constraint that the outer commutator is nontrivial). For each quadruple, we compute $\widehat{D}_2$ in exact rational arithmetic and evaluate $\|\widehat{D}_2\|_{\mathrm{F}}^2$. The worst case over all 405 quadruples is 67586345/67108864. The computation yields 16 distinct Frobenius norms, ranging from $82431849/1073741824 \approx 0.077$ to the worst case above. □

4.4 The Simplex Obstruction

Despite the Frobenius-norm contraction at level 2, the operator norm $\|\widehat{D}_2\|_{\mathrm{op}}$ equals 1 for 270 of the 405 valid quadruples, blocking a naive submultiplicativity argument for higher levels. The following observation explains the structure of this obstruction.

Observation 4.6 (Simplex Structure)**.** *The* 270 *quadruples with* $\|\widehat{D}_2\|_{\mathrm{op}} = 1$ *each preserve exactly one direction in* $\mathbb{R}^4$ *(as the eigenvector of* $\widehat{D}_2^\top \widehat{D}_2$ *with eigenvalue* 1*). These directions are precisely the projections of the* 5 *standard basis vectors* $e_0, \ldots, e_4 \in \mathbb{R}^5$ *onto the hyperplane* $V = \{\sum v_i = 0\}$*:*

$$v_k = \frac{e_k - \bar{e}}{\|e_k - \bar{e}\|}, \quad \bar{e} = \tfrac{1}{5}(1,1,1,1,1)^\top, \quad k = 0, \ldots, 4.$$

These 5 *directions form a* ***regular simplex*** *in* $\mathbb{R}^4$*, with pairwise inner products* $|\langle v_i, v_j\rangle| = 1/4$ *for all* $i \neq j$*. Each direction is preserved by exactly* 54 *quadruples (perfectly symmetric).*

The regularity of this structure — five directions, uniform distribution, constant pairwise angle — suggests a general phenomenon rather than an artifact of S_5.

Theorem 4.7 (Dimensional obstruction for S_n, $n \geq 6$)**.** *For* S_n *with* $n \geq 5$*, the level-*1 *Fourier transform* $D_1 = (P_\sigma P_\tau)^2$ *in the standard representation* $(n-1, 1)$ *has singular values* $\underbrace{1, \ldots, 1}_{n-3}$, $1/8$, 0*. In particular,* D_1 *preserves a subspace of dimension* $n-3$ *isometrically.*

For $n \geq 6$*, the minimum intersection of two such preserved subspaces in* $\mathbb{R}^{n-1}$ *is* $2(n-3) - (n-1) = n-5 \geq 1$*, so every level-*2 *quadruple has* $\|\widehat{D}_2\|_{\mathrm{op}} = 1$ *and level-*3 *contraction fails. Consequently, the spectral contraction mechanism of this paper is specific to* S_5*.*

Proof. The projections $P_\sigma = I_V - v_\sigma v_\sigma^\top$ and $P_\tau = I_V - v_\tau v_\tau^\top$ share the property that $\langle v_\sigma, v_\tau\rangle = -1/2$ for all adjacent pairs (independent of n). Decomposing $V = W \oplus W^\perp$ where $W = \mathrm{span}(v_\sigma, v_\tau)$ (dimension 2): on $W^\perp$ (dimension $n-3$), both projections act as the identity, giving $n-3$ singular values equal to 1; on W, the 2×2 block $(P_\sigma P_\tau)^2\big|_W$ has $D_1^\top D_1\big|_W$ with eigenvalues $1/64$ and 0 (trace $= 4/256$, determinant $= 0$), giving singular values $1/8$ and 0. The 2×2 block is independent of n.

For $n \geq 6$: two subspaces of dimension $n-3$ in $\mathbb{R}^{n-1}$ intersect in dimension $\geq n-5 \geq 1$. Any unit vector in this intersection is preserved isometrically by both level-1 blocks, so $\|\widehat{D}_2\|_{\mathrm{op}} = 1$ for every valid quadruple. □

Remark 4.8. *The simplex structure (Theorem 4.6) does not generalize to* S_n *for* $n \geq 6$*: exhaustive computation confirms that all* 1620 *valid quadruples for* S_6 *and all* 4725 *for* S_7 *have operator norm* 1 *(versus* $270/405 = 66.7\%$ *for* S_5*), and the preserved directions do not form a regular simplex. The dimensional obstruction explains this:* S_5 *is the unique case where the preserved subspace dimension* $(n-3 = 2)$ *allows disjoint placement in the ambient space* $(n-1 = 4)$*.*

4.5 Theorem 3: Level 2 to Level 3 Contraction

Lemma 4.9 (Direction Mixing)**.** *For every valid level-*3 *configuration* $(\mathrm{quad}_A, \mathrm{quad}_B)$ *in which both sub-blocks have* $\|\widehat{D}_2\|_{\mathrm{op}} = 1$*, the two sub-blocks preserve* ***different*** *directions.*

Proof. We establish two claims.

Claim 1: A level-2 quadruple $(\sigma_1, \tau_1, \sigma_2, \tau_2)$ with $\|\widehat{D}_2\|_{\mathrm{op}} = 1$ preserves direction v_k if and only if all four transpositions fix element k.

The forward direction is immediate: if all transpositions fix k, then $\langle v_k, v_{(a,b)}\rangle = (\delta_{k,a} - \delta_{k,b})/(\sqrt{2}\,\|e_k - \bar{e}\|) = 0$ for each null vector $v_{(a,b)}$, so $v_k \in W_A^\perp \cap W_B^\perp$ and $\widehat{D}_2 v_k = v_k$. For the backward direction, if $\|\widehat{D}_2 v_k\|_{\mathrm{op}} = 1$ and $\widehat{D}_2 = D_{1,A} D_{1,B} D_{1,A}^\top D_{1,B}^\top$ with each factor having operator norm ≤ 1, the chain of norm inequalities forces v_k into the preserved subspace $W^\perp$ of each level-1 block, hence $\langle v_k, v_{(a,b)}\rangle = 0$ for all four transpositions, i.e., all fix k.

Claim 2: If both quadruples have all transpositions fixing element k, the level-3 commutator is trivial.

All eight transpositions lie in $\mathrm{Stab}(k) \cong S_4$, which is *solvable* with derived series $S_4 \rhd A_4 \rhd V_4 \rhd \{e\}$, where $V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ is the Klein four-group. Following the Barrington commutator tree within $\mathrm{Stab}(k)$: level-1 targets (commutators of adjacent transpositions) are 3-cycles in $[S_4, S_4] = A_4$; level-2 targets (commutators of 3-cycles) lie in $[A_4, A_4] = V_4$; the level-3 commutator $[\alpha_A, \alpha_B]$ with $\alpha_A, \alpha_B \in V_4$ lies in $[V_4, V_4] = \{e\}$, since V_4 is abelian.

This contradicts the requirement $[\alpha_A, \alpha_B] \neq e$ for a valid level-3 configuration. Therefore both quadruples cannot preserve the same direction. □

Theorem 4.10 (Level-3 Contraction)**.** *For every valid level-3 configuration:*

$$\|\widehat{D}_3\|_{\mathrm{op}} \leq \lambda := 0.0354.$$

In particular, $\|\widehat{D}_3\|_{\mathrm{op}} < 1$ *for all* 73810 *configurations.*

Proof. For each valid level-3 pair $(\mathrm{quad}_A, \mathrm{quad}_B)$, we compute

$$\widehat{D}_3 = \widehat{D}_{2,A} \cdot \widehat{D}_{2,B} \cdot \widehat{D}_{2,A}^\top \cdot \widehat{D}_{2,B}^\top$$

and evaluate the operator norm (maximum singular value) numerically. The maximum over all 73810 configurations is $\lambda \approx 0.0353$, strictly less than 1.

The contraction occurs because:

- If at least one sub-block has $\|\widehat{D}_2\|_{\mathrm{op}} < 1$, submultiplicativity gives $\|\widehat{D}_3\|_{\mathrm{op}} \leq \|\widehat{D}_{2,A}\|_{\mathrm{op}}^2 \cdot \|\widehat{D}_{2,B}\|_{\mathrm{op}}^2 < 1$.
- If both sub-blocks have $\|\widehat{D}_2\|_{\mathrm{op}} = 1$ but preserve different directions $v_A \neq v_B$ (guaranteed by Theorem 4.9), then $\widehat{D}_{2,A}$ maps v_B to a vector with component $|\langle v_A, v_B\rangle| = 1/4$ along v_A, causing the product to contract. □

4.6 Theorem 4: Induction to All Levels

Theorem 4.11 (Exponential Contraction)**.** *For all Barrington branching programs for point functions* CC_T *over* $\{0,1\}^\ell$ *with* $\ell = 2^d$*,* $d \geq 3$*:*

$$\|\widehat{D}_d\big((4,1)\big)\|_{\mathrm{op}} \leq \lambda^{4^{d-3}},$$

where $\lambda = 0.0354$*.*

Proof. At depth $d+1$, the commutator structure gives $\widehat{D}_{d+1} = \widehat{D}_A \cdot \widehat{D}_B \cdot \widehat{D}_A^\top \cdot \widehat{D}_B^\top$. By submultiplicativity of the operator norm and $\|M^\top\|_{\mathrm{op}} = \|M\|_{\mathrm{op}}$:

$$\|\widehat{D}_{d+1}\|_{\mathrm{op}} \leq \|\widehat{D}_A\|_{\mathrm{op}} \cdot \|\widehat{D}_B\|_{\mathrm{op}} \cdot \|\widehat{D}_A^\top\|_{\mathrm{op}} \cdot \|\widehat{D}_B^\top\|_{\mathrm{op}} = \|\widehat{D}_d\|_{\mathrm{op}}^4.$$

The base case $\|\widehat{D}_3\|_{\mathrm{op}} \leq \lambda$ is Theorem 4.10. Iterating: $\|\widehat{D}_d\|_{\mathrm{op}} \leq \lambda^{4^{d-3}}$. □

5 Main Result

Theorem 5.1 (Main Theorem)**.** *For all Barrington branching programs for point functions* CC_T *over* $\{0,1\}^\ell$ *with* $\ell = 2^d$*,* $d \geq 3$*:*

(a) $D_T = D_{T'}$ *for all* $T, T' \in \{0,1\}^\ell$*.*

(b) $d_{\mathrm{TV}}(D_T, \mathrm{Uniform}(S_5)) \leq 6 \cdot \lambda^{4^{d-3}}$*, where* $\lambda = 0.0354$*.*

(c) In particular, $d_{\mathrm{TV}}(D_T, U) = \mathrm{negl}(\ell)$*.*

Proof. Part (a) is Theorem 3.1. For part (b), by Theorem 2.7:

$$d_{\mathrm{TV}}(D_T, U)^2 \leq \frac{1}{4} \sum_{\rho \neq \mathrm{triv}} d_\rho \cdot \|\widehat{D}(\rho)\|_{\mathrm{F}}^2.$$

By Theorem 4.1, the sign representation contributes zero. For each remaining representation ρ of dimension d_ρ, Theorem 4.2 gives a base case $\|\widehat{D}_3(\rho)\|_{\mathrm{op}} \leq \lambda_\rho < 1$ for all valid level-3 configurations. The submultiplicativity argument of Theorem 4.11 applies to each ρ independently (it uses only $\|M^\top\|_{\mathrm{op}} = \|M\|_{\mathrm{op}}$ and submultiplicativity, which hold in any representation), yielding $\|\widehat{D}_d(\rho)\|_{\mathrm{op}} \leq \lambda_\rho^{4^{d-3}}$. Since $\|M\|_{\mathrm{F}}^2 \leq d_\rho \cdot \|M\|_{\mathrm{op}}^2$:

$$\sum_\rho d_\rho \cdot \|\widehat{D}(\rho)\|_{\mathrm{F}}^2 \leq \sum_\rho d_\rho^2 \cdot \lambda_\rho^{2 \cdot 4^{d-3}} \leq \Big(\sum_\rho d_\rho^2\Big) \cdot \lambda^{2 \cdot 4^{d-3}},$$

where $\lambda = \max_\rho \lambda_\rho = \lambda_{(4,1)} \approx 0.0353$ is the bottleneck (Theorem 4.2). Since $\sum_\rho d_\rho^2 = 4^2 + 5^2 + 6^2 + 5^2 + 4^2 = 118$:

$$d_{\mathrm{TV}}(D_T, U)^2 \leq \frac{118}{4} \cdot \lambda^{2 \cdot 4^{d-3}}, \qquad d_{\mathrm{TV}}(D_T, U) \leq \sqrt{\tfrac{118}{4}} \cdot \lambda^{4^{d-3}} < 6 \cdot \lambda^{4^{d-3}}.$$

For part (c), since $\lambda < 1$ and 4^{d-3} grows super-exponentially in $d = \log_2 \ell$, the bound is negligible in ℓ. □

For concrete values:

ℓ	Depth d	$\|\widehat{D}_{(4,1)}\|_{\mathrm{op}}$ bound	d_{TV} bound
8	3	3.54×10^{-2}	2.12×10^{-1}
16	4	1.57×10^{-6}	9.42×10^{-6}
32	5	6.08×10^{-24}	3.65×10^{-23}
64	6	1.37×10^{-93}	8.21×10^{-93}

6 Extensions and Open Problems

6.1 Conjunctions with Wildcards

A conjunction $\mathrm{CC}_{T,W}$ with target $T \in \{0,1\}^\ell$ and wildcard set $W \subseteq [\ell]$ outputs 1 iff $x_i = T_i$ for all $i \notin W$. In the branching program, wildcard positions output σ_i for both bit values. Obfuscation of conjunctions has been studied extensively [4, 5, 6]; our results provide an information-theoretic complement to the computational security guarantees in that literature.

Theorem 3.1 extends directly to conjunctions: $D_{T,W} = D_{T',W}$ for all T, T' with the same wildcard set W. The spectral contraction depends on the wildcard density $|W|/\ell$. Computational evidence shows strong contraction for $|W|/\ell \leq 1/2$, weakening contraction for $|W|/\ell = 3/4$, and no contraction for $|W|/\ell = 1$ (the trivial function). The precise threshold is an open problem.

6.2 Beyond Point Functions

For general branching programs computing the same Boolean function via different constructions, "strong universal mixing" ($D_{C_0} = D_{C_1}$) is **false**: programs with different transposition assignments produce different distributions. However, "weak universal mixing" (both distributions converge to uniform independently) appears to hold: computational evidence shows equivalent circuits with different pair assignments achieving similar d_{TV} values. Proving this would strengthen the security guarantee for indistinguishability obfuscation of general NC^1 functions.

Computational experiments reveal a stronger pattern. For $\ell = 4$, two programs computing AND_4 with different transposition pairs yield Fourier norms $\|\widehat{D}(4,1)\|_{\mathrm{F}}^2 = 0.147$ and 0.293 respectively — clearly different distributions — yet *identical* total variation distances $d_{\mathrm{TV}} = 0.309$ to uniform. The same equidistance persists for OR-of-AND circuits ($d_{\mathrm{TV}} \approx 0.19$ for two structurally different implementations). If this "equidistance" phenomenon is general, it would mean that $d_{\mathrm{TV}}(D_C, U)$ depends only on the Boolean function computed, not on the specific Barrington construction — a property strictly stronger than weak universal mixing.

6.3 Generalization to Other Groups

Theorem 4.7 shows that our level-3 contraction mechanism does not extend to S_n for $n \geq 6$. The dimensional obstruction is fundamental: in the standard representation of S_n, the level-1 Fourier transform preserves $n-3$ of $n-1$ dimensions, and the ratio $(n-3)/(n-1) \to 1$ as $n \to \infty$, leaving insufficient room for contraction.

This does not rule out uniform mixing for S_n with $n > 5$ via alternative proof strategies. Two natural directions remain open:

1. *Higher-level contraction.* Although level-3 contraction fails for $n \geq 6$, it is conceivable that contraction occurs at level 4 or higher, where the commutator structure may break the shared preserved directions. Computational verification of this hypothesis for S_6 is feasible but requires enumerating $O(10^{12})$ level-4 configurations.
2. *Non-standard representations.* For S_n with $n > 5$, non-standard representations (e.g., $(n-2, 2)$ or $(n-2, 1, 1)$) may exhibit better contraction properties. The bottleneck analysis (Theorem 4.2) would need to be repeated for each n.

For the matrix groups $\mathrm{GL}_n(\mathbb{Z}_q)$ used in modern lattice-based iO constructions [3, 19, 13], the representation theory is richer (principal series, cuspidal, Steinberg representations), and the dimensional obstruction may not apply in the same form. Determining whether spectral contraction is achievable for $\mathrm{GL}_2(\mathbb{F}_q)$ with small q remains a concrete open problem, especially timely given recent impossibility results for evasive LWE [9].

6.4 Why S_5 is Structurally Unique

The results of this paper reveal a precise structural explanation for why S_5 is the unique symmetric group where level-3 spectral contraction succeeds. Three properties conspire:

1. S_5 **is non-solvable.** A_5 is simple, so iterated commutators of non-identity elements never reach the identity — this is what makes Barrington's theorem work over S_5.
2. $\mathrm{Stab}(k) \cong S_4$ **is solvable with derived length 3.** The derived series $S_4 \rhd A_4 \rhd V_4 \rhd \{e\}$ terminates at depth 3, which is precisely the level of the commutator tree where contraction occurs. This forces the direction-mixing mechanism (Theorem 4.9).
3. **The preserved subspace dimension equals half the ambient dimension.** D_1 preserves $n-3 = 2$ of $n-1 = 4$ dimensions, allowing two preserved subspaces to be placed in general

position (intersection dimension = 0). For $n \geq 6$, the ratio $(n-3)/(n-1) > 1/2$ makes this impossible (Theorem 4.7).

The coincidence of these three properties — non-solvability of S_5, solvability of S_4 at derived length exactly 3, and the critical dimension ratio — makes S_5 the unique locus where this proof strategy succeeds.

7 Computational Verification

The proofs of Theorems 4.4 and 4.5 use exact rational arithmetic (implemented in `sympy`) with a verified orthonormal basis for the standard representation. Theorem 4.10 uses numerical computation (`numpy`) over all 73810 valid configurations.

The full verification suite consists of 15 scripts, publicly available at:

https://github.com/Jefemaestro33/barrington-mixing

Script	Content	Method
01	Ambivalence of S_5 (similarity blindness)	Exhaustive
02	Collision analysis, $\ell = 4$	Exhaustive
03	Collision scaling, $\ell = 4 \to 8$	Sampled
04	Fourier spectral analysis	Exact + sampled
05	Barrington construction + Kilian obfuscation	Exhaustive
06	Contraction operator (unconditional distribution)	Analytic
07	T-independence proof verification	Exhaustive
08	Conjunctions with wildcards	Exhaustive
09	Universal mixing investigation	Exact + sampled
10	Complete spectral contraction proof	Exact + numerical
11	Eigenvector analysis (simplex discovery)	Numerical
12	Level-3 exhaustive verification	Numerical
13	All-representations contraction verification	Numerical
14	S_6/S_7 dimensional obstruction (Theorem 4.7)	Exact + exhaustive
15	Algebraic direction mixing verification (Theorem 4.9)	Exact

Scripts 10–13 collectively verify the main theorem in under 20 seconds on a standard laptop. Script 13 verifies Theorem 4.2: operator-norm contraction for all five non-trivial, non-sign representations across all 73810 level-3 configurations. Script 14 verifies Theorem 4.7: the dimensional obstruction for S_6 (1620 quadruples, all with operator norm 1) and S_7 (4725 quadruples), confirming that level-3 contraction fails for $n \geq 6$. Script 15 verifies each step of the algebraic proof of Theorem 4.9: the equivalence between preserved directions and fixed elements, the derived series of S_4, and the triviality of $[V_4, V_4]$. Scripts 01–09 provide additional cross-validation and extensions.

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